Tutorial - 1 03/01/2023

(1)
$$2x - 3y + z = 1$$
$$x + 2y - 3z = 4$$
$$4x - y - 2z = 8$$
$$5x + y - 13z = 4$$

Sol^n: The corresponding augmented matrix is

$$\left[\begin{array}{ccc|c} 2 & -3 & 1 & 1 \\ 1 & 2 & -3 & 4 \\ 4 & -1 & -2 & 8 \\ 5 & 1 & -13 & 4 \end{array}\right] \underset{\sim}{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 2 & -3 & 1 & 1 \\ 4 & -1 & -2 & 8 \\ 5 & 1 & -13 & 4 \end{array}\right] \underset{\sim}{\begin{array}{l} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 4R_1 \\ R_4 = R_4 - 5R_1 \end{array}}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & -7 & 7 & -7 \\ 0 & 5 & -4 & 6 \\ 0 & -9 & 2 & -16 \end{array}\right] \underset{\sim}{-\frac{1}{7}R_2} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 5 & -4 & 6 \\ 0 & -9 & 2 & -16 \end{array}\right] \underset{\sim}{\begin{array}{l} R_3 = R_3 - 5R_2 \\ R_4 = R_4 + 9R_2 \end{array}}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -7 & -7 \end{array}\right] \underset{\sim}{R_4 = R_4 + 7R_3} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The reduced system of equations is

$$\left.\begin{array}{r} x + 2y - 3z = 4 \\ y - z = 1 \\ z = 1 \\ 0 = 0 \end{array}\right\} \quad \begin{array}{l} \Rightarrow z = 1 \\ y - z = 1 \Rightarrow y = 2 \\ x + 2y - 3z = 4 = x + 4 - 3 = 4 \Rightarrow x = 3 \end{array}$$

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(2) Find the REF of $\begin{bmatrix} 1 & 2 & -5 & 2 \\ 3 & -1 & 2 & 4 \\ 4 & 1 & -3 & 6 \\ 2 & -3 & 7 & 2 \end{bmatrix}$

Sol^n:

$$\begin{bmatrix} 1 & 2 & -5 & 2 \\ 3 & -1 & 2 & 4 \\ 4 & 1 & -3 & 6 \\ 2 & -3 & 7 & 2 \end{bmatrix} \xrightarrow[R_4 = R_4 - 2R_1]{\substack{R_2 = R_2 - 3R_1 \\ R_3 = R_3 - 4R_1}} \begin{bmatrix} 1 & 2 & -5 & 2 \\ 0 & -7 & 17 & -2 \\ 0 & -7 & 17 & -2 \\ 0 & -7 & 17 & -2 \end{bmatrix}$$

$$\xrightarrow[R_4 = R_4 - R_2]{R_3 = R_3 - R_2} \begin{bmatrix} 1 & 2 & -5 & 2 \\ 0 & -7 & 17 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(3)

Find the REF of $\begin{bmatrix} 1 & 2 & 0 & 3 \\ 4 & 5 & 0 & 6 \\ 7 & 8 & 0 & 9 \end{bmatrix}$

Solⁿ: $\begin{bmatrix} 1 & 2 & 0 & 3 \\ 4 & 5 & 4 & 6 \\ 7 & 8 & 8 & 9 \end{bmatrix} \underset{R_3 = R_3 - 7R_1}{\overset{R_2 = R_2 - 4R_1}{\sim}} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & -3 & 4 & -6 \\ 0 & -6 & 8 & -12 \end{bmatrix}$

$\underset{\sim}{R_3 = R_3 - 2R_2} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & -3 & 4 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

(4) Solve the system of equations by Gauss elimination method

$2x + 3y = 18$

$5x - 2y = 7$

$-3x + 4y = 7$

Solⁿ. The corresponding augmented matrix is

$\left[\begin{array}{cc|c} 2 & 3 & 18 \\ 5 & -2 & 7 \\ -3 & 4 & 7 \end{array}\right] \underset{R_3 = R_3 + \frac{3}{2}R_1}{\overset{R_2 = R_2 - \frac{5}{2}R_1}{\sim}} \left[\begin{array}{cc|c} 2 & 3 & 18 \\ 0 & -\frac{19}{2} & -38 \\ 0 & \frac{17}{2} & 34 \end{array}\right] \underset{\sim}{R_3 = R_3 + \frac{17}{19}R_2}$

$\left[\begin{array}{cc|c} 2 & 3 & 18 \\ 0 & -\frac{19}{2} & -38 \\ 0 & 0 & 0 \end{array}\right]$

The reduced system of equations is

$\therefore \left.\begin{array}{l} 2x + 3y = 18 \\ -\frac{19}{2}y = -38 \\ 0 = 0 \end{array}\right\} \Rightarrow y = 4$

$\therefore 2x + 3y = 18$

$\Rightarrow 2x + 12 = 18 \Rightarrow x = 3$

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